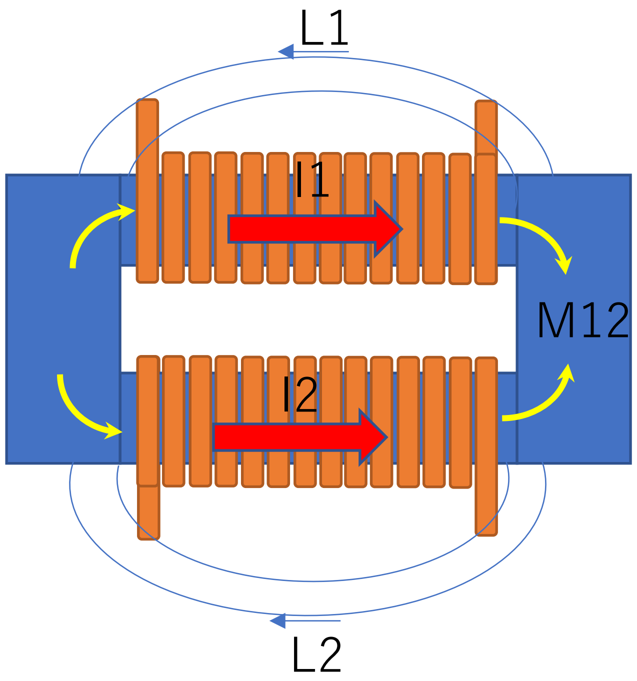
Instruction Manual for the High-Speed Calculation Tool for Magnetically Coupled DC-DC Converters



Objective:
For reactors that utilize spatial magnetic flux leakage, it is difficult to determine their characteristics through magnetic circuit calculations. Therefore, the aim is to establish a method that enables operational calculations by using 3D FEM to pre-determine the reactor's nonlinear magnetic characteristics based on the current state of each coil.

Conclusion:
A method to achieve the objective was established using the following three steps (circa November 2015):

STEP 1:
Maxwell 3D magnetostatic analysis script. Calculation of self-inductance, mutual inductance, and leakage flux at specific current levels.

STEP 2:
Design of Experiments (DOE) using Python; sequentially obtaining calculation results while varying the current values used in the script above. Creation of 2D maps for L and M based on currents Ia and Ib.

STEP 3:
Using the maps above, Python is used to calculate current and flux ripple waveforms based on operating conditions (taking a few seconds per operating condition).

Ansys Maxwell 3D script(VBS)

Set analysis conditions, Ia, Ib, and component dimensions as parameters. Automatically generate the model via the program and execute the calculation. Output the results in text format.(CSV)

C: > Users > kazi3 > Documents > my_program > Github > Maxwell1 > maxwellのpython > github > github_REACTOR_TRY_25model_analysis.VBS

This document contains many non-basic ASCII unicode characters Disable Non ASCII Highlight

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'2015/10/28　磁気結合型コンバータ向けﾘｱｸﾄﾙのインダクタンス計算スクリプト　コイルA電流,コイルB電流->La,Lb,Mabを算出 磁性材の非線形性を考慮

dim Turn,Cuh,Cut,Cucl,Cur,Clcnt,Clw,Clh,Icorew,Icoreh,Cll,Uwplus,Uhplus,Udepth,Usidew,Ang,Corecl  'プリミティブな値。これらからモデルが決まる。
dim Cur1,Cur2 '右、左コイルの電流　各々プラスで磁界が打ち消す方向に電流が流れるようにモデルを組む
dim leak1x,leak1z,leak2x,leak2z '漏れ磁束計測用の座標　１がコイル1で、2がコイル2

dim Startedge,Icorecl,Uinw,Length,Ilength,Holelength,Coilw,Coilh,Uw,Uh  '1次的に計算される寸法
dim pi
pi=3.14159265358979323846264338327950288419716939993
dim ii
dim str '文字列の入れ物
dim savepath  'ファイルのセーブするパス＆ファイル名
dim savedata  '出力する自己インダクタンス　相互インダクタンス算出結果のパス＆ファイル
dim saveflux  '磁束密度を出力するパス
'-----------------ここが編集する部分--------------------------------------------
savepath="D:\Userarea\J0125789\Documents\ANSYS_CAE\ANSOFT\MAXWELL\MAXWELL16_SCRIPT\25MODEL_DETAILED_COIL\Project1.aedt" 'ファイルのセーブするパス
savedata="D:\Userarea\J0125789\Documents\ANSYS_CAE\ANSOFT\MAXWELL\MAXWELL16_SCRIPT\25MODEL_DETAILED_COIL\Project1_Maxwell3DDesign1.txt" '出力する
saveflux="D:\Userarea\J0125789\Documents\ANSYS_CAE\ANSOFT\MAXWELL\MAXWELL16_SCRIPT\25MODEL_DETAILED_COIL\"  '磁束密度を記録するpath

Cur1 = 100.0  'とりあえず100A
Cur2 = 50.0 'とりあえず50A

Turn=32 ':ターン数（片側コイル）
Cuh=6.5 ':銅線幅
Cut=1.8 ':銅線厚み
Cucl=0.4  'コイル間のクリアランス
Cur=5 ':コイル内径r
Clcnt=4 ':コイル間中央の隙間
Clw=3.05  ':コアコイル横側クリアランス
Clh=3.25  ':コアコイル縦側クリアランス
Icorew=32 ':Iコアの幅
Icoreh=25 ':Iコアの高さ
Cll=3.56  ':コイル前後のコアとの距離
Uwplus=0.6  ':Uコア幅のIコア幅に対する差分
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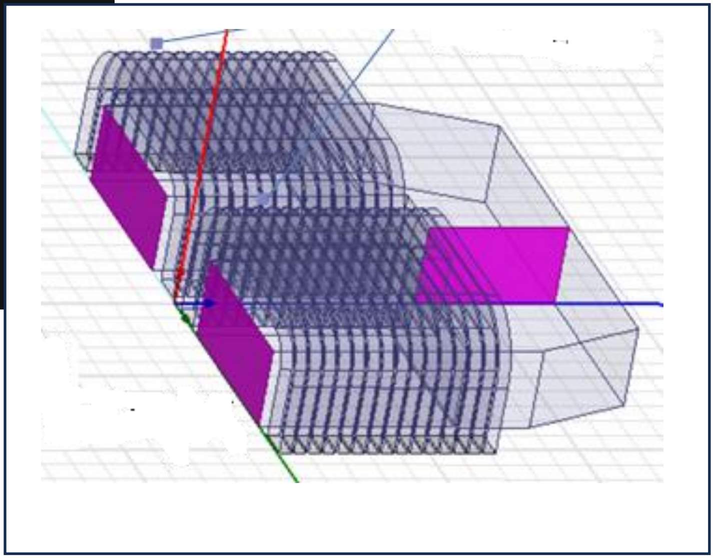
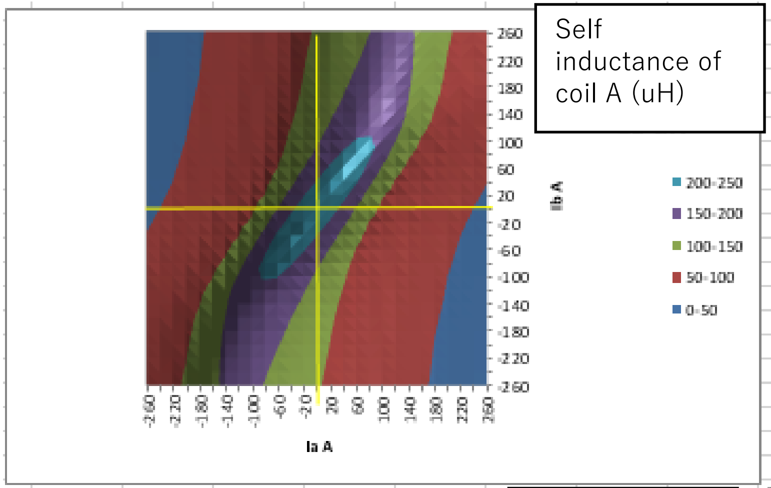


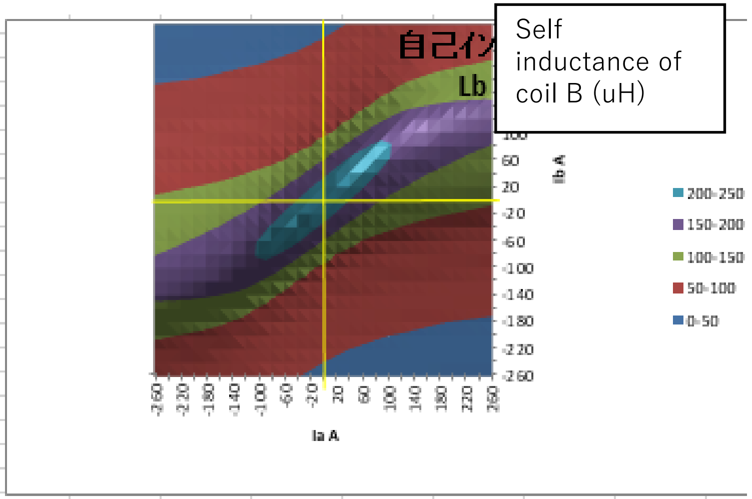
Such models are automatically generated based on parameters, and the calculations run automatically.

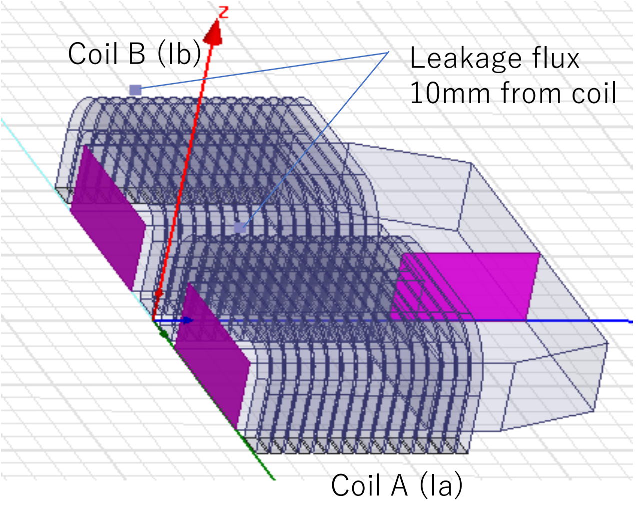
Python DOE script

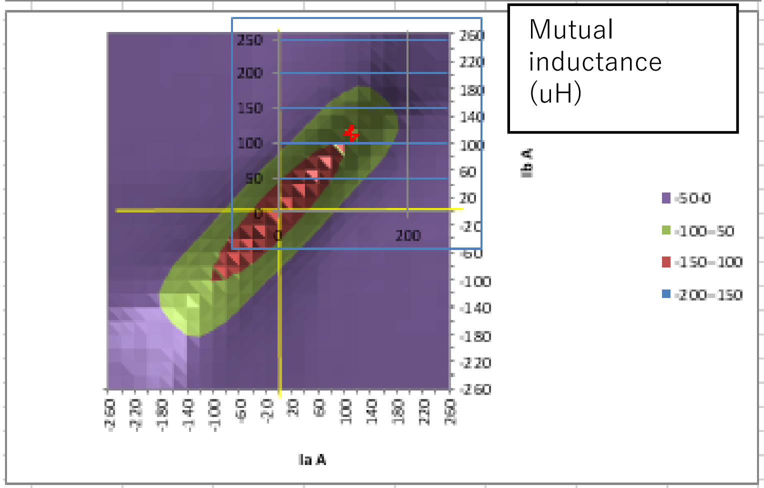
Result example

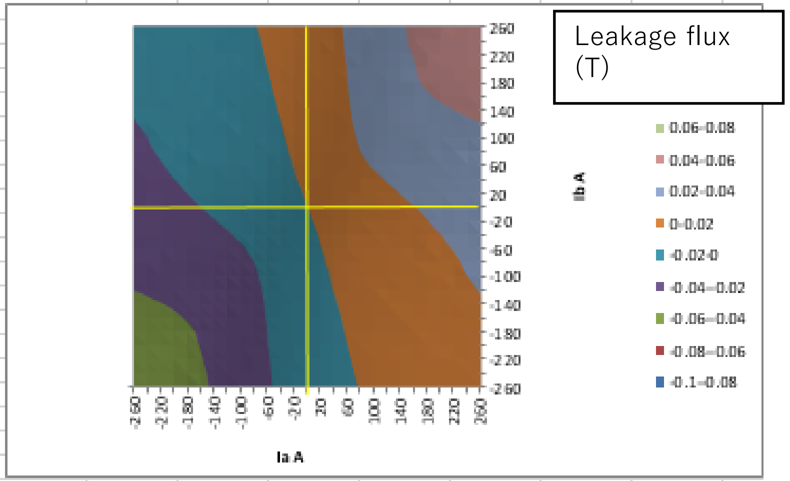
By using Maxwell script on previous page, create L and M maps based on the current conditions of the two coils.







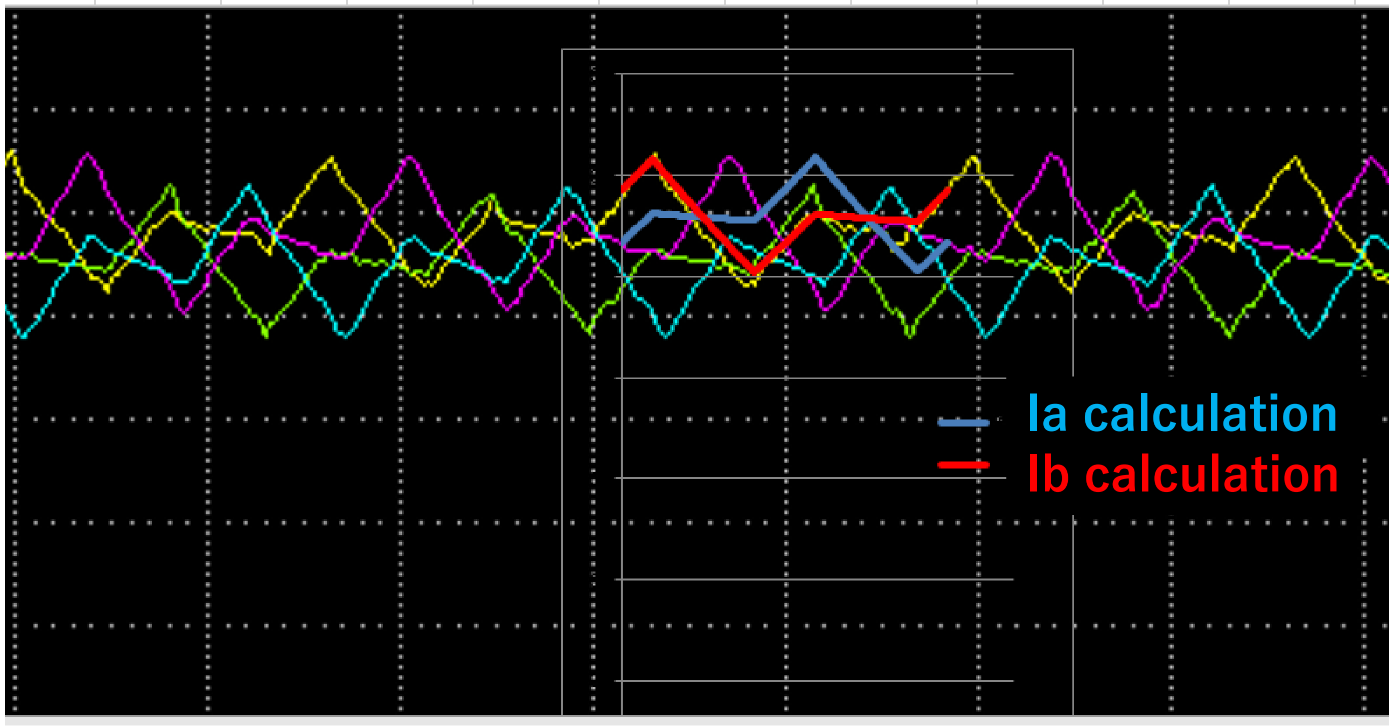




Calculation of ripple current and flux waveform with Python

Calculations are performed based on the constraint VL = -dϕ/dt and a pre-generated L-M map.

Example of calculated current results (superimposed with measured values)



The calculated current waveform matches the measured data with relatively high accuracy; therefore, it is concluded that this method is fully capable of supporting the design evaluation of magnetically coupled reactors.